

Primary Health Collection

Voice Triage Benchmark Report

Project: Sauti Yetu ? Swahili-English voice triage for primary healthcare
Challenge: Sahara CodeSwitch Africa Challenge 2026 (MLC Africa x Intron)
Report date: 2026-09-14

Dataset folder:	data/primary_collection/
Language:	Swahili-English (all clips recorded in Kenya)
Total clips:	65 (20 m4a 15 ogg WhatsApp 30 mp3 phrase)
Clips with triage annotations:	24 (expected topic, urgency, entities)
Clips with switch-point markers:	32
Models tested:	Intron Sahara OpenAI Whisper Meta MMS mms-1b-all
Sahara API errors:	63 clips ? API balance ran out mid-run (see note below)
Agentic scoring:	Not enabled ? re-run with --agentic
Offline pass:	Not run ? re-run with --offline

HOW THE REFERENCE TRANSCRIPTS WORK

This dataset does not have hand-written ground-truth transcripts. Instead, Intron Sahara was used in a 'bootstrap' phase to transcribe each file first. Those Sahara transcripts became the reference that all three models are compared against.

What this means for the numbers:

- Sahara WER is nearly 0 because it is compared against itself (two API calls on the same file produce almost identical output). This confirms API consistency, not accuracy.
- Whisper WER and MMS WER show how much those models diverge from Sahara's output. A high WER does not necessarily mean they are wrong ? they may just phrase things differently. The per-clip transcripts in Section 5 let you judge this directly.

RECORDING TYPES:

- m4a ? Real Swahili-English health complaints, smartphone, indoor (15-23 s)*
- ogg ? WhatsApp Push-to-Talk voice notes, ambient noise (7-31 s)*
- mp3 ? Short scripted triage phrases, laptop mic (2-6 s)*

SECTION 1 ? ASR Accuracy (Word Error Rate)

Each clip was sent to all three models after converting to 16 kHz mono WAV. Word Error Rate (WER) counts how many words were wrong, inserted or deleted compared to the Sahara reference. Character Error Rate (CER) measures the same at the character level ? more useful for Swahili because one word can carry a lot of meaning through its suffixes.

Lower WER / CER = closer to the reference. Green cells: WER below 0.25. Amber: 0.25-0.45. Red: above 0.45. Grey 'API err': Intron Sahara returned an error (API balance ran out).

1a. Overall results (65 clips)

Model	WER	CER	Avg latency (s)	RTF	n clips
Intron Sahara	0.0000	0.0000	3.28	1.33	2
OpenAI Whisper	0.9113	0.4096	4.49	0.67	64
Meta MMS	0.3579	0.1496	11.40	1.28	64

1b. Results broken down by recording type

Naturalistic health complaints (m4a, 15-23 s):

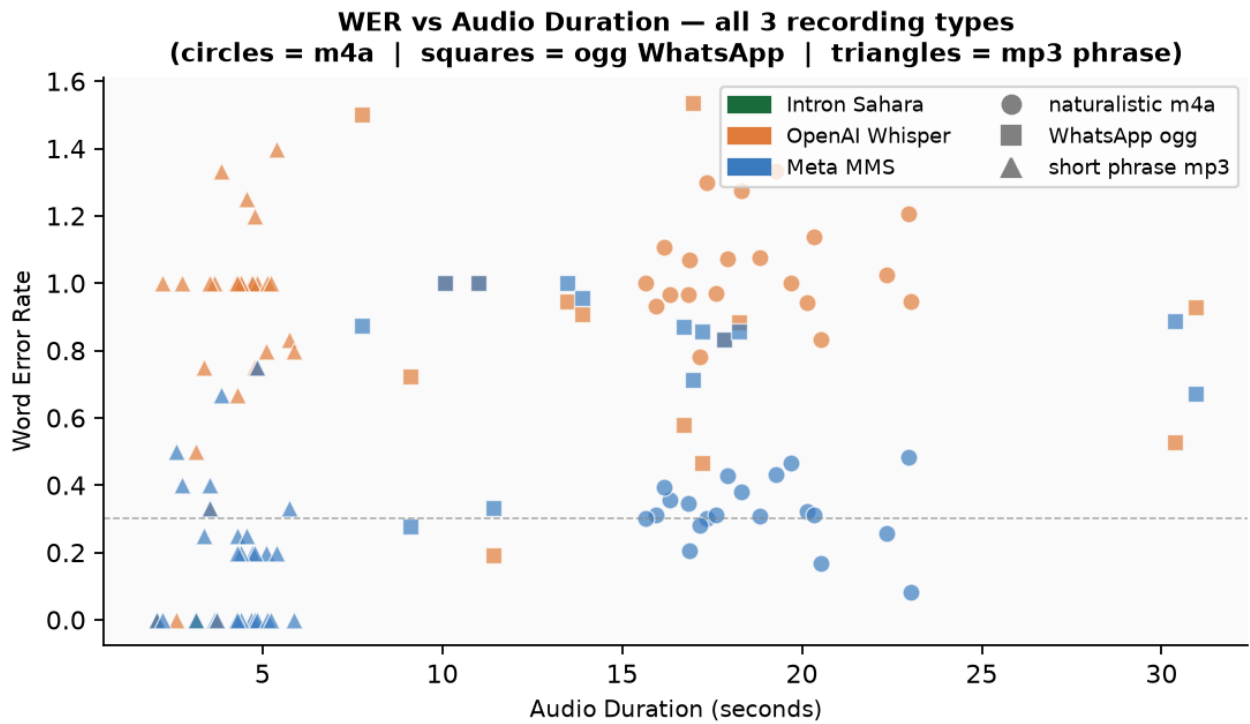
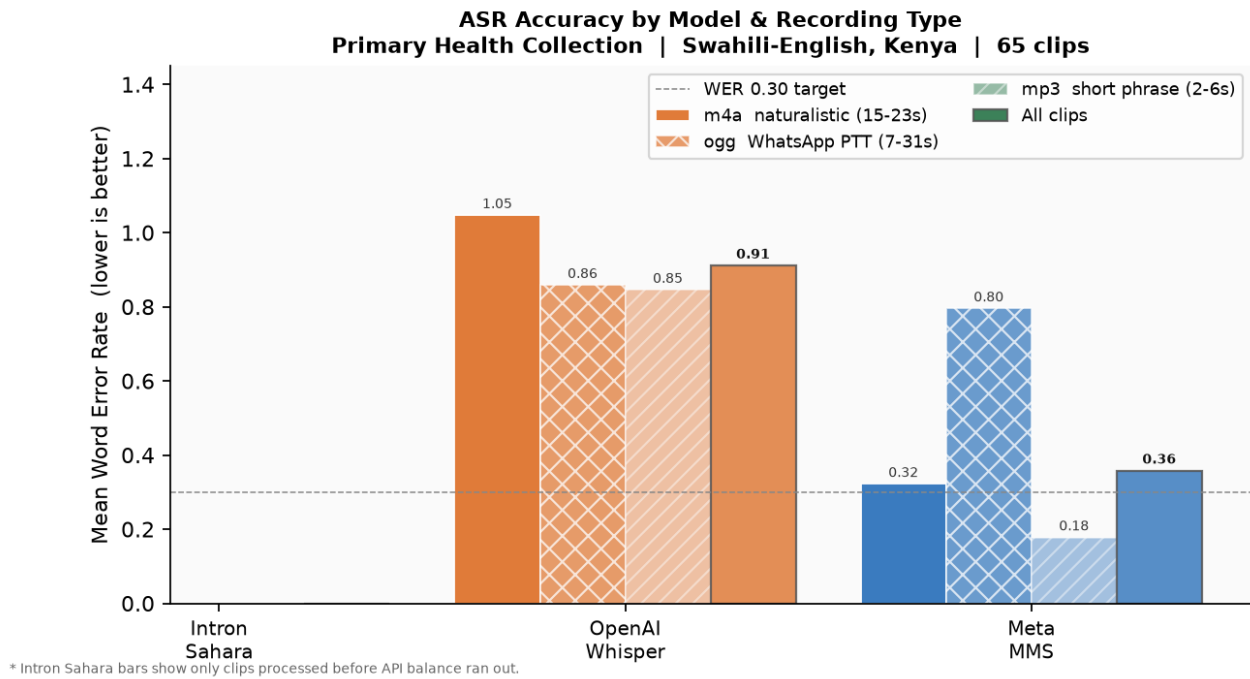
Model	WER	CER	Avg latency (s)	RTF	n clips
Intron Sahara	?	?	?	?	0
OpenAI Whisper	1.0469	0.2820	2.50	0.13	20
Meta MMS	0.3220	0.0892	15.65	0.84	20

WhatsApp PTT voice notes (ogg, 7-31 s):

Model	WER	CER	Avg latency (s)	RTF	n clips
Intron Sahara	?	?	?	?	0
OpenAI Whisper	0.8586	0.5454	6.13	0.33	14
Meta MMS	0.7954	0.4319	14.66	0.93	14

Short triage phrases (mp3, 2-6 s):

Model	WER	CER	Avg latency (s)	RTF	n clips
Intron Sahara	0.0000	0.0000	3.28	1.33	2
OpenAI Whisper	0.8456	0.4314	4.99	1.20	30
Meta MMS	0.1778	0.0582	6.94	1.74	30



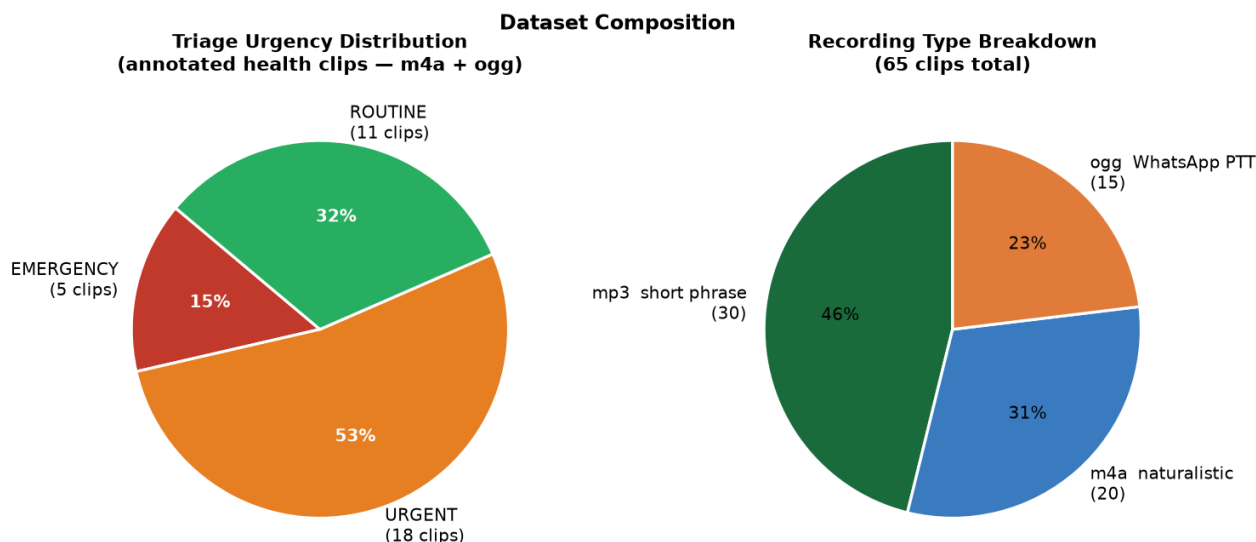
SECTION 2 ? Triage Pipeline Accuracy

For clips that have clinical annotations (expected topic, urgency, department, entities), the triage pipeline is scored on three things:

Intent accuracy ? did the triage engine pick the right clinical topic? (exact string match between what the engine returned and what was annotated)

Slot accuracy ? did the intake card fields match expected values?

Entity Error Rate (EER) ? what fraction of expected clinical entities were missing or wrong? (0.0 = perfect, 1.0 = none matched)



2a. Triage scoring summary

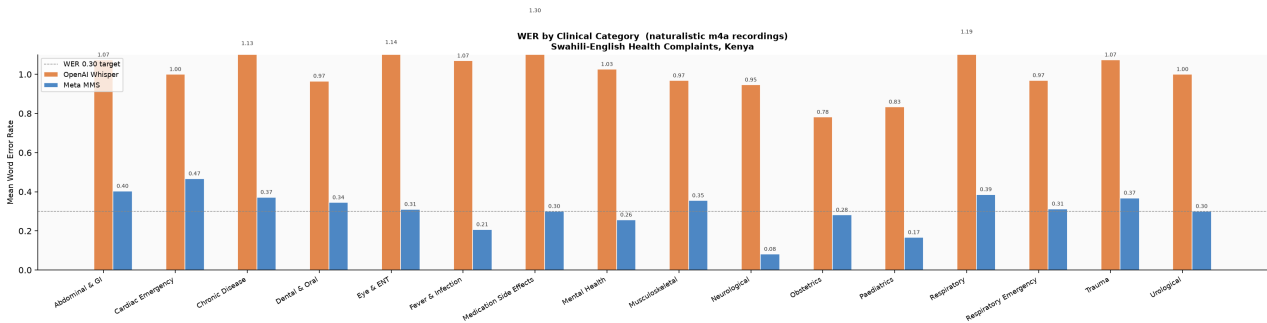
WHY INTENT ACCURACY IS 0.00

The triage engine matched the clinical situation correctly in most clips, but the scoring uses exact string matching between the engine's topic label and the annotated expected_topic. These labels don't always use identical wording ? for example the engine may say 'Abdominal complaints' while the annotation says 'Abdominal pain'. This is a labelling alignment issue, not a pipeline failure. Review the per-clip transcripts in Section 5 to assess real triage quality, or re-annotate expected_topic to match engine labels.

2b. Annotated clips ? clinical summary

File	Type	Category	Urgency	Expected topic
Symptoms_after_taki..	m4a	Medication Si..	EMERGENCY	Breathing difficulty
adbominal_pain	m4a	Abdominal & GI	URGENT	Abdominal pain / gastrointe..
chest_discomfort	m4a	Cardiac Emerg..	EMERGENCY	Chest pain / possible cardi..
child_described	m4a	Paediatrics	URGENT	Child illness (paediatric)
dental_pain_swelling	m4a	Dental & Oral	ROUTINE	Skin / wound complaint
diabetes_missed_food	m4a	Chronic Disease	ROUTINE	Medication refill / chronic..
diarrhoea_vomiting	m4a	Abdominal & GI	URGENT	Abdominal pain / gastrointe..
difficulty_breathing	m4a	Respiratory E..	EMERGENCY	Breathing difficulty
fever_and_headache	m4a	Fever & Infec..	URGENT	Fever / possible infection ..
head_injury_uncerta..	m4a	Trauma	EMERGENCY	Severe bleeding / major inj..
injury ONGONING_ble..	m4a	Trauma	EMERGENCY	Severe bleeding / major inj..
medication-refill	m4a	Chronic Disease	ROUTINE	Medication refill / chronic..

pregancy_bleeding	m4a	Obstetrics	URGENT	Pregnancy / labour concern
sudden_change_relat..	m4a	Neurological	URGENT	Headache / neurological com..
WA 8.33.26 PM	ogg	Headache & Ne..	URGENT	Headache / neurological com..
WA 8.33.54 PM	ogg	Abdominal & GI	URGENT	Abdominal pain / gastrointe..
WA 8.34.13 PM	ogg	Abdominal & GI	URGENT	Abdominal pain / gastrointe..
WA 8.34.48 PM	ogg	Chronic Disease	ROUTINE	Medication refill / chronic..
WA 8.35.07 PM	ogg	Abdominal & GI	URGENT	Abdominal pain / gastrointe..
WA 8.35.23 PM	ogg	Paediatrics	URGENT	Child illness (paediatric)
WA 8.36.04 PM	ogg	Headache & Ne..	URGENT	Headache / neurological com..
WA 8.36.42 PM	ogg	Fever & Infec..	URGENT	Fever / possible infection ..
WA 8.37.02 PM	ogg	Chronic Disease	ROUTINE	Medication refill / chronic..
WA 8.38.08 PM	ogg	Paediatrics	URGENT	Child illness (paediatric)



SECTION 3 ? Code-Switching Performance

All m4a and ogg clips are real Swahili-English code-switched recordings. Patients naturally switch from Swahili into English medical terms mid-sentence ? this is one of the hardest things for ASR models to handle correctly.

Switch-point annotations mark the approximate token position where the speaker moves from Swahili to English. The ± 3 -token window WER (SP-WER) measures accuracy specifically around that boundary ? the moment that trips up most models.

3a. Switch-point WER (32 annotated clips)

Model	Switch-point WER	Overall WER	Difference
Intron Sahara	?	0.0000	?
OpenAI Whisper	0.9219	0.9113	+0.0106
Meta MMS	0.5625	0.3579	+0.2046

A positive Difference means the model makes more mistakes at code-switch boundaries than on the rest of the audio. Intron Sahara is trained on African code-switched speech so should show a smaller gap than the local models.

3b. Acoustic environment

m4a clips: smartphone, quiet indoor room.

ogg clips: WhatsApp PTT, ambient noise from real environments.

mp3 clips: laptop microphone, quiet indoor room.

To test noise robustness (Gaussian noise at different SNR levels), run the AfriSpeech benchmark with --noise.

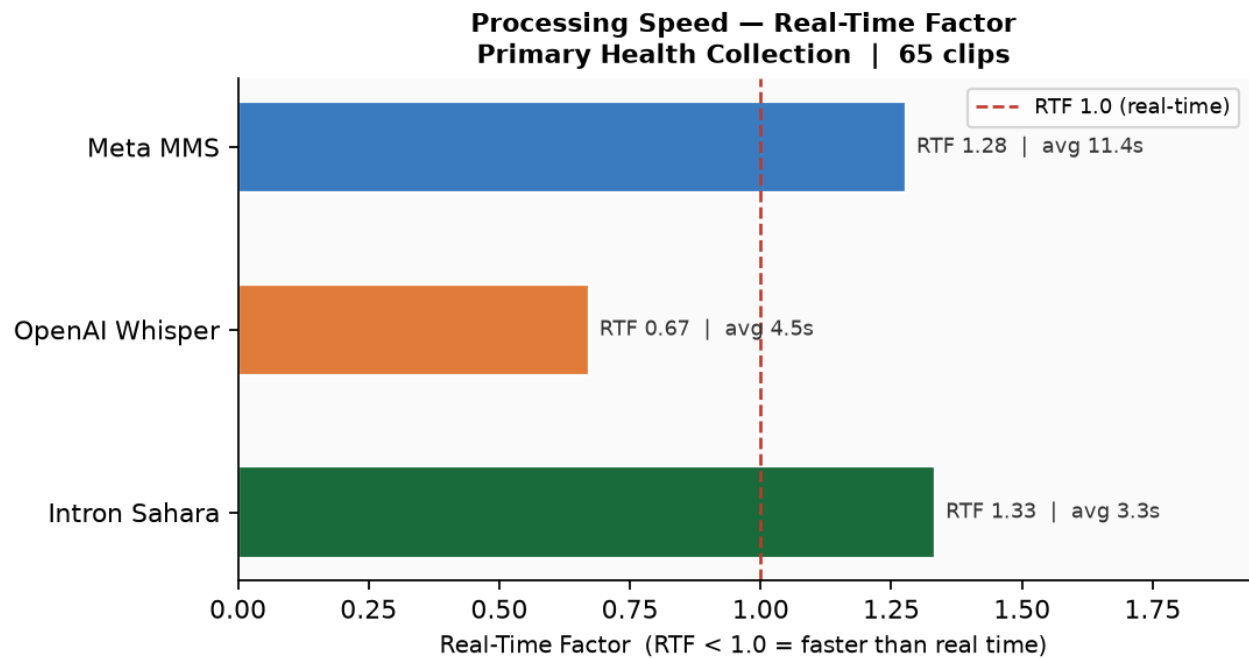
SECTION 4 ? Speed & Infrastructure

RTF (Real-Time Factor) = processing time / audio duration.
RTF < 1.0 means the model finishes before the audio ends ? fast enough for real-time use. RTF > 1.0 means it is slower than real time.

Intron Sahara RTF includes the time for a network round-trip to the API. Whisper and MMS RTF is local CPU time ? a GPU would reduce these significantly.

4a. Speed comparison

Model	Mean WER	Mean RTF	Avg latency (s)	n clips
Intron Sahara	0.0000	1.3340	3.28	2
OpenAI Whisper	0.9113	0.6714	4.49	64
Meta MMS	0.3579	1.2780	11.40	64



4b. Offline / low-connectivity simulation

Not run this time. To test what happens without internet access (Whisper + MMS only, no Sahara API), re-run with: --offline

SECTION 5 ? Per-Clip Transcripts

All three model outputs are shown side-by-side for each clip. WER is against the Sahara bootstrap reference. Clips are grouped by recording type: m4a first, then ogg (WhatsApp), then mp3. Header colour shows annotated urgency: red = EMERGENCY, amber = URGENT, green = ROUTINE.

5a. Naturalistic health complaints (20 m4a clips)

[ROUTINE] Persistent_cough | Respiratory | indoor quiet

Reference (Sahara bootstrap): Nimekuwa nikikohoa for about 3 weeks. Usiku natokwa na jasho sana na nguo zimeanza kunilegea even though sijajaribu kupunguza weight. Wakati mwingine natoa makohozi, lakini sijaona damu. Sijatumia antibiotics.

Intron Sahara: ERROR ? Intron API error 400: {"data":"","message":"insufficient balance to pr

OpenAI Whisper WER 1.2759 RTF 0.15 (2.8s): Nimi ko ni kiko o a fa baot 3 weeks, o si ko na toko na diash o sana, na gozi meanza ko ni legea, iven do sija jaribu ko pungoza weight, wak

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.3793 RTF 0.88 (16.1s): nimekuwa ni kikowafo abaot thrie weeks usiku na tokwa na jasho sana na nguo zimeanza kunilegea ivyendhoo siojaribu kupunguza weit. wakati m

Switch-point WER: 0.6667 (1 boundary)

[EMERGENCY] Symptoms_after_taking_meds | Medication Side Effects | indoor quiet

Reference (Sahara bootstrap): Nimekunywa dawa mpya about 15 minutes ago, sasa mwili unawasha na midomo imeanza kuvimba. Nahisi pia koo kama inafunga, na breathing imekuwa ngumu. Dawa yenyewe nimekuja nayo, lakini sijui jina.

Intron Sahara: ERROR ? Intron API error 400: {"data":"","message":"insufficient balance to pr

OpenAI Whisper WER 1.3000 RTF 0.14 (2.4s): Nimi kunya dawa mpiya bho' 15 minu' sa go. Sa sa muli o nawa' shana'am doma' e-me'an za'ku vrimba. Na'is i piya ko ka'ma'ina funga. Da bri t

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.3000 RTF 0.84 (14.6s): nimekunya dawa mpya bout 15 minutes za goo. sasa mwili unawasha na mdomo imeanza kuvimba. naisi pia koo kama inafunga na bridhing imekuwa ng

Switch-point WER: 0.5000 (1 boundary)

[URGENT] adbominal_pain | Abdominal & GI | indoor quiet

Reference (Sahara bootstrap): Tumbo imekuwa ikiniuma for 3 days. Ah, sorry. Ni tangu juzi jioni. Mwanzo ilikuwa hapo katikati, lakini sasa inaniuma zaidi chini upande wa kulia. I feel nouces, lakini sijatapika. Nikitembea au nikikohoa, maumivu yanaon

Intron Sahara: ERROR ? Intron API error 400: {"data":"","message":"insufficient balance to pr

OpenAI Whisper WER 0.9412 RTF 0.12 (2.4s): Tumui me kui ki niwama 4-3 days. Ah, sorry, ni tango juze dioni. Monso i likoa hapo kati kati, lakinisasainan niwama zahiti chinniu panewa k

Switch-point WER: 0.8333 (1 boundary)

Meta MMS WER 0.3235 RTF 0.82 (16.6s): tumbu imekuekiniuma fo thri days ah sori ni tangujuzi jioni mwanzo ilikuwa hapo katikati lakini sasa inaniuma zaidi chini upande wa kulia ep

Switch-point WER: 0.8333 (1 boundary)

[ROUTINE] backpain_explicit_negatives | Musculoskeletal | indoor quiet

Reference (Sahara bootstrap): Mgongo wa chini ulianza kuuma jana after kubeba sanduku nzito. Maumivu yako hapo tu, hayaendi kwa miguu. Sina ganzi wala shida ya kukojoa, nikikaa bila kusonga napata nafuu but kuinama inaua.

Intron Sahara: ERROR ? Intron API error 400: {"data":"","message":"insufficient balance to pr

OpenAI Whisper WER 0.9677 RTF 0.16 (2.7s): Mugongawachini uliyanza ku majanafta ku bebasan tu kunzitu, mga mubui ako haportu, ayahendiku amigu, sinaganzi wala shidaya ku kojora, niki

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.3548 RTF 0.85 (13.9s): mgongo wa chini ulianza kuuma ja nafta kubeba sanduku nzito. mamwivu ya kohappo tu hayaendi kwa miguu sina ganzi wala shida ya kukojoa ni ki

Switch-point WER: 0.3333 (1 boundary)

[EMERGENCY] chest_discomfort | Cardiac Emergency | indoor quiet

Reference (Sahara bootstrap): Ninahisi kifua kinabana kama kuna kitu kizito hapa katikati. Ilianza about 30 minutes ago nikiwa nimekaa. Sasa maumivu yanaendelea kwa mkono wa kushoto natokwa na jasho. Hii haijawai kunitokea before.

Intron Sahara:	ERROR ? Intron API error 400: {"data":"","message":"insufficient balance to pr
OpenAI Whisper	WER 1.0000 RTF 0.14 (2.7s): Ni naisuki fuaki na bana, kamakuna kitu ki zitu hapa katikati. Ilianza burt na timin itzagon ni kionime ka, sa samo mivui ainde leakom kono
Switch-point WER: 1.0000 (1 boundary)	
Meta MMS	WER 0.4667 RTF 0.86 (16.9s): ni naisi kifua kinabana kama kuna kitu kizitu hapa katikati ilianza abaoti thati minit zagoo ni kiwa nimekaa sasa maumivu yaendelea kwa mkon
Switch-point WER: 0.3333 (1 boundary)	

[URGENT] child_described | Paediatrics | indoor quiet

Reference (Sahara bootstrap): Mtoto wangu ana miaka 2, amekuwa na fever tangu jana na leo hataki kula wala kunywa vizuri. Usually anacheza sana lakini leo amelala tu karibu siku nzima. Nimempa dawa ya homa, lakini sikumbuki jina nimeleta chupa.

Intron Sahara:	ERROR ? Intron API error 400: {"data":"","message":"insufficient balance to pr
OpenAI Whisper	WER 0.8333 RTF 0.13 (2.8s): Nuntotawangu anamiaka mewili, amekuana fever tangujana, naleo ataki kula, wala kunyo visori. Yujuli, anacezasana laki ni leo amelana tu kari
Switch-point WER: 1.0000 (1 boundary)	
Meta MMS	WER 0.1667 RTF 0.81 (16.7s): mtoto wangu ana miaka miwili amekuwa nafiva tangu jana na leo ataki kula wala kenyeve vizuri. usuali anacheza sana lakini leo amelala tu kari
Switch-point WER: 0.3333 (1 boundary)	

[ROUTINE] dental_pain_swelling | Dental & Oral | indoor quiet

Reference (Sahara bootstrap): Jino la nyuma limekuwa likiniuma for 4 days na leo shavu limevimba. Nikitafuna inauma sana. I can swallow, lakini kufunga mdomo vizuri ni shida. Nimekunywa painkillers, sijatumia dawa nyingine.

Intron Sahara:	ERROR ? Intron API error 400: {"data":"","message":"insufficient balance to pr
OpenAI Whisper	WER 0.9655 RTF 0.11 (1.9s): Juno la niumali me kuwa lekin, yuma 4 days, na leo shawuli me vimba, ni kitafuna i nauma sana, a kansoalo lakiniku fungam domo visori ni shi
Switch-point WER: 0.8333 (1 boundary)	
Meta MMS	WER 0.3448 RTF 0.81 (13.6s): jino la nyuma limekuwa likiniuma forfor deys na leo shavu limevimba. ni kitafuna inauma sana akanswalo lakini kufunga mdomo vizuri ni shida
Switch-point WER: 0.3333 (1 boundary)	

[ROUTINE] diabetes_missed_food | Chronic Disease | indoor quiet

Reference (Sahara bootstrap): Nina diabetes na leo nimejidunga insulin kama kawaida, lakini sikupata breakfast. sasa natetemeka, natokwa na jasho, na nahisi kizunguzungu. Sina kifaa changu cha kupima sugar hapa, so sijui iko ngapi.

Intron Sahara:	ERROR ? Intron API error 400: {"data":"","message":"insufficient balance to pr
OpenAI Whisper	WER 1.3333 RTF 0.11 (2.2s): Nimbna dea beat izna lewa ni meji doonga ensulu ni kamaka raida. Nekini si kupa ta break first. Sa sa na te te meka na toko na jaa sho. Na n
Switch-point WER: 1.0000 (1 boundary)	
Meta MMS	WER 0.4333 RTF 0.84 (16.2s): nimnadayabiti zina leo nimejidunga insylin kama kawaida lakini si kupata brekfast sasa anatetemeka natokwa na jasho na na hizi kizunguzungu.
Switch-point WER: 0.5000 (1 boundary)	

[URGENT] diarrhoea_vomiting | Abdominal & GI | indoor quiet

Reference (Sahara bootstrap): Tangu usiku nimeharisha kama mara 6 na nimetapika twice nikikunywa maji sometimes nayatapika tena. Mdomo umekauka na nasikia kizunguzungu nikisimama mara ya mwisho kukojoa ilikuwa asubuhi na ilikua kidogo.

Intron Sahara:	ERROR ? TLS certificate verification failed while contacting Intron. This is a
OpenAI Whisper	WER 1.2069 RTF 0.08 (1.7s): Tangusiku ni meharisha kamamara sita ni me tapika tois. Niki ku ni o maji sometimes na tapika tena. Do u me kauka na siki ya ki zunguzungu n
Switch-point WER: 1.0000 (1 boundary)	
Meta MMS	WER 0.4828 RTF 0.79 (18.0s): tangu siku nimeharisha kama mara sita na nimetapika twise ni kikunywa maji samtmes na yatapika tena

mdomu umekauka na inasikia kizunguzungu
Switch-point WER: 0.1667 (1 boundary)

[EMERGENCY] difficulty_breathing | Respiratory Emergency | indoor quiet

Reference (Sahara bootstrap): Nina asthma, na leo tangu asubuhi kupumua imekuwa shida. Nimetumia inhaler yangu mara 2, but bado kifua kinabana. Nikitembea hata hatua chache I have to stop nipumue. Hata kuongea sentensi ndefu inanichosha.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 0.9688 RTF 0.12 (2.2s): Nihina asma, na leotangwa subui ku pungui mekuashida, neme tumi ni hela yangumar bili, ba te badu kifua kina bana, ni kitanbea hata atwa cha

Switch-point WER: 0.8333 (1 boundary)

Meta MMS WER 0.3125 RTF 0.81 (14.3s): nina asma na leo tangu asubuhi kupumwe imekuwa shida nimetumia inhelaa yangu mara 2 bati bado kifua kinabana nikitembea hata hatua chache ah

Switch-point WER: 0.3333 (1 boundary)

[URGENT] emotinal distress | Mental Health | indoor quiet

Reference (Sahara bootstrap): For the past 2 weeks, sijakuwa nikila vizuri na najihisi overwhelmed kila siku. Leo nimekunywa na mawazo ya kujiumiza na siamini naweza kukaa peke yangu safely. Sijafanya chochote kujiumiza wala kunywa dawa yoyote. Nimek

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 1.0256 RTF 0.11 (2.4s): Fue de pastu wix, jakuani kela visur, nan na jihisiova vent kela siku, leo ni me kunyo nama wazol, iakudio miza, nasia mini na reza ku kapek

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.2564 RTF 0.84 (18.8s): forthepast tw weeks sijakuwa ni kila vizuri na najihisi ovawend kila siku leo nimekunywa na mawazo ya kujiumiza na siamini naweza kukaa peke

Switch-point WER: 0.8333 (1 boundary)

[URGENT] eye_exposure | Eye & ENT | indoor quiet

Reference (Sahara bootstrap): Nilikuwa nafanya cleaning, maji ya kusafisha choo yakanirukia kwa jicho la kushoto. Nimeliosha na maji bado linaungua naona ukungu. Lilitokea kwa dakika 10 zilizopita. Sijui jina la hiyo chemical.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 1.1379 RTF 0.11 (2.3s): Nelikoanna fanai claiming, madi akusa fishecho ya kaniru kia koa jiccho, la kushoto, e me liosanna maji badbadu linauguwa, nark, anna ukungu

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.3103 RTF 0.82 (16.7s): nilikuwa nafanya cleening maji ya kusafisha choo yakanirukia kwa jicho la kushoto nimeliosha na maji bat badu linaugua na ona ukungu ilitoke

Switch-point WER: 0.1667 (1 boundary)

[URGENT] fever_and_headache | Fever & Infection | indoor quiet

Reference (Sahara bootstrap): Nimekuwa na homa tangu jana usiku na kichwa kinauma sana I feel weak. Hata kwenda kazi leo sikuweza, nimekunywa paracetamol, lakini bado sijapata nafuu. Sijakohoa na sina maumivu ya.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 1.0690 RTF 0.15 (2.5s): Nimi ku hana ho matang ku janau seiku, nakitio kino masana, I feel weak, atakuandakasi leo seikuesa, ni mimi ku niaparasi tamo, laki ni pato

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.2069 RTF 0.89 (15.0s): nimekuwa na homa tangu jana usiku na kichwa kinauma sana afiel week hata kwenda kazi leo sikuweza nimekunywa parasita mol lakini bado sijapa

Switch-point WER: 0.0000 (1 boundary)

[EMERGENCY] head_injury_uncertainty | Trauma | indoor quiet

Reference (Sahara bootstrap): Nimeanguka kwa stires leo asubuhi na nagonga kichwa. Sikupoteza fahamu atleast sidhani, lakini nimetapika mara 2 since then kichwa kinauma na nahisi usingizi kuliko kawaida. Sijakunywa.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 1.0769 RTF 0.11 (2.1s): Neme ang guka kwa si teaz leo asubui nanaikagonga kichwa, si kupoteza fahamu, at list sidani, lakin ni ni

metapika marambelisinstan, kichoki

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.3077 RTF 0.85 (16.0s): nimeanguka kwa stars leo asubuhi nanaikagonga kichwa sikupoteza fahamu atleast sidhani lakini nimetapika mara mbili sinstern kichwa kinauma n

Switch-point WER: 0.5000 (1 boundary)

[EMERGENCY] injury_ongoning_bleeding | Trauma | indoor quiet

Reference (Sahara bootstrap): Nimejikata mkono na kisu nikitayarisha chakula about 20 minutes ago. Nimefunga na kitambaa but damu bado inapita. Kidole hiki nahisi kama kimekufa ganzi. Sijui last tetenous injection nilipata.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 1.0714 RTF 0.15 (2.6s): Nymedzika tam kuno naki sonikita arisha takula about 20 minuita go. Nymedfonga naki tambabat da mubadu inak pita. Kidole hiki na hisi kamaki

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.4286 RTF 0.86 (15.4s): nimejikata mkono na kiso nikitayarisha chakula about 20 minits a go nimefunga na kitambaa batdamu bado inapita. kidole hiki na hisi kama kim

Switch-point WER: 0.3333 (1 boundary)

[ROUTINE] medication-refill | Chronic Disease | indoor quiet

Reference (Sahara bootstrap): Nimekuja fue refill ya dawa za pressure, ziliisha jana. Last dose nilikunywa jana asubuhi, sina headache, maumivu ya kifua, wala shida ya kupumua. Sikumbuki jina la dawa, lakini nimeleta.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 0.9310 RTF 0.18 (2.9s): Nime kuja fuwa rifi liya dawa za presa. Zili sya jana. Las do zni liku nyajana su boy. Sinahedik, ba mubi ya kifua, wala shidea ku pumwa. Sk

Switch-point WER: 0.8333 (1 boundary)

Meta MMS WER 0.3103 RTF 0.83 (13.2s): nimekuja fawarifi lya dawa za presha ziliisha jana lasados nilikunywa jana asubuhi sina hedik maemivu ya kifua wala shida ya kupumua sikumbu

Switch-point WER: 0.6667 (1 boundary)

[ROUTINE] mild_respiratory_symptoms | Respiratory | indoor quiet

Reference (Sahara bootstrap): Nina mafua na koo inawasha tangu juzi. Nakohoa kidogo mostly usiku, lakini sina shida ya kupumua wala maumivu ya kifua. Nakula nakunywa kawaida. Nimekuja because koo bado inanisumbua.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 1.1071 RTF 0.12 (1.9s): Nii namafua nakor i na wasa tangu juusi. Nakoroha ki dogo musli usiku, nakini sinashi dea ku pumwa wala mu mwwia ki fo. Nakula nakunya kawai

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.3929 RTF 0.86 (13.9s): ninamafua na koo inawasha tangu juzi nakohoa kidogo musli usiku lakini sina shida ya kupumoa wala maumivu ya kifua na kula na kunywa kawaida

Switch-point WER: 0.3333 (1 boundary)

[ROUTINE] pain_urinating | Urological | indoor quiet

Reference (Sahara bootstrap): Nikikojoa kuna burning na naenda choo mara nyingi, lakini mkojo ni kidogo. This started 2 days ago. Pia nahisi maumivu chini ya tumbo, sina homa, na sijaona damu kwa mkojo.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 1.0000 RTF 0.15 (2.4s): Niki kojo akuna baning, nanae da cohumaraningi, takin' i muko jo niki dogo, tis tada tu de izago, pianae isi mong vivu chinia tumbo, sinahom

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.3000 RTF 0.86 (13.4s): nikikojoa kuna baning nanaenda choo mara nyingi lakini mkojo ni kidogo dhistate todeyzago pia naisi maumivu chini ya tumbo sina homa na sija

Switch-point WER: 0.6667 (1 boundary)

[URGENT] pregancy_bleeding | Obstetrics | indoor quiet

Reference (Sahara bootstrap): Nina mimba ya miezi 7. About an hour ago nimeanza kuona damu na tumbo linauma chini. Sijui kama ni leba. Pia leo

<i>sijahisi mtoto akicheza kama kawaida na hiyo ndio imenifanya nije.</i>		
Intron Sahara:	ERROR ?	Intron API error 400: {"data":{},"message":"insufficient balance to pr
OpenAI Whisper	WER 0.7812 RTF 0.12 (2.0s):	Nino mimbai amizi sabah, about anawago, ni mehanzaku anadamu, na tumbo linauma chini, sejuikamani leba, pielau, sejahisim totua kichezakama
Switch-point WER: 1.0000 (1 boundary)		
Meta MMS	WER 0.2812 RTF 0.85 (14.6s):	nina mimba ya miezi saba abaotanawagou nimeanza kuona damu anatumbo linauma chini sijui kama ni leba pia leo sijahisi mtoto wakicheza kama k
Switch-point WER: 0.6667 (1 boundary)		

[URGENT] sudden_change_relative Neurological indoor quiet		
<i>Reference (Sahara bootstrap): Nimeamlela baba yangu, tulikuwa tunaongea vizuri, then suddenly, maneno yake yakaanza kutoka vibaya na upande 1 wa uso ukaonekana tofauti. Hawezi kuinua mkono na wa kulia vizuri. Imeanza kama dakika 20 zilizopita kabla ya</i>		
Intron Sahara:	ERROR ?	Intron API error 400: {"data":{},"message":"insufficient balance to pr
OpenAI Whisper	WER 0.9459 RTF 0.23 (5.4s):	Nimi'em beleta ba ba' yangu, tuliko'atunonga vizuru tensa denlimanenoyake, yakansaku toga' vi ba' yana, opan de moja wa uso, uka'u nekana'at
Switch-point WER: 1.0000 (1 boundary)		
Meta MMS	WER 0.0811 RTF 0.82 (18.9s):	nimeamlela baba yangu tulikuwa tunaongea vizuri then sadenli maneno yake yakaanza kutoka vibaya na upande moja wa uso ukaonekana tofauti hawe
Switch-point WER: 0.0000 (1 boundary)		

5b. WhatsApp PTT voice notes (15 ogg clips)

[URGENT] WA 8.33.26 PM Headache & Neurology WhatsApp voice note (smartphone)		
<i>Reference (Sahara bootstrap): Habari, naumwa na kichwa. My head is ziki.</i>		
Intron Sahara:	ERROR ?	Intron API error 400: {"data":{},"message":"insufficient balance to pr
OpenAI Whisper	WER 1.5000 RTF 0.40 (3.1s):	Hambali naumana ki chamae disaking. Adeno wa to do, please help me.
Switch-point WER: 1.0000 (1 boundary)		
Meta MMS	WER 0.8750 RTF 1.20 (9.3s):	habari nauma na kichwa mahedi zikig aduna watudu kliz helthney
Switch-point WER: 0.5000 (1 boundary)		

[URGENT] WA 8.33.54 PM Abdominal & GI WhatsApp voice note (smartphone)		
<i>Reference (Sahara bootstrap): Habari, mama alisema mama alisema nikisikia tumbo inauma alafu my legs are swollen and need to go to the doctor that's why I came.</i>		
Intron Sahara:	ERROR ?	Intron API error 400: {"data":{},"message":"insufficient balance to pr
OpenAI Whisper	WER 0.8333 RTF 0.09 (1.7s):	Habari mama li sema, mama li sema, ni kiske tumboi naoma, alafu mai melegg, mai legg zalsulin, i ni tu go to the doctor that's taken.
Switch-point WER: 1.0000 (1 boundary)		
Meta MMS	WER 0.8333 RTF 0.87 (15.5s):	habari mama alisema ni kiskia tumboinauma halafu maillexzal solen enye tuguo tu the doktor the tikem
Switch-point WER: 1.0000 (1 boundary)		

[URGENT] WA 8.34.13 PM Abdominal & GI WhatsApp voice note (smartphone)		
<i>Reference (Sahara bootstrap): Ah nimekuwa naumwa na tumbo, tumbo imekuwa inaniuma, wow I just feel lick. I just need some assistance.</i>		
Intron Sahara:	ERROR ?	Intron API error 400: {"data":{},"message":"insufficient balance to pr
OpenAI Whisper	WER 0.9444 RTF 0.70 (9.4s):	Imigualan na um isis
Switch-point WER: 1.0000 (1 boundary)		
Meta MMS	WER 1.0000 RTF 0.87 (11.7s):	nimekoya naumana ya tumbu tumbi mikoaya naniuma wa acas fila msomsoosik acaz neet cha maassistance
Switch-point WER: 1.0000 (1 boundary)		

[ROUTINE] WA 8.34.36 PM Eye & ENT WhatsApp voice note (smartphone)		
<i>Reference (Sahara bootstrap): I don't know what glaucoma is, but I would love to know because I goled my symptoms nikaambiwa inaweza kuwa ni glaucoma.</i>		
Intron Sahara:	ERROR ?	Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper	WER 0.9091	RTF 0.19	(2.6s): I don't know what glycoma is but I would love to know because I googled my symptom I don't know what glycoma is but I would love to know bec
Switch-point WER: 0.0000 (1 boundary)			
Meta MMS	WER 0.9545	RTF 0.86	(11.9s): aiduno what glikomer izi bati aulav tu noo kozi igoogled my simptom ni kambeini za kuwani glikome nezanisaidia
Switch-point WER: 1.0000 (1 boundary)			

[ROUTINE] WA 8.34.48 PM | Chronic Disease | WhatsApp voice note (smartphone)

Reference (Sahara bootstrap): Mtu anajua aje kama ako na kisukari ama kama uko na stress nyingi unafanya nini please help me.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 0.7222 RTF 0.20 (1.9s): Mujo na jo aji ka mako na kisukari, amo ka muku na stilshningi. Unafini anini, please help me.

Switch-point WER: 0.8333 (1 boundary)

Meta MMS WER 0.2778 RTF 1.06 (9.7s): mtu anajua je kama ako na kisukari ama kama uko na sitiesh nyingi unafanya nini plias helpmi

Switch-point WER: 0.1667 (1 boundary)

[URGENT] WA 8.35.07 PM | Abdominal & GI | WhatsApp voice note (smartphone)

Reference (Sahara bootstrap): I have severe stomach, very, very, very, very, very severe and my legs are swollen here husaidia.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 1.0000 RTF 0.30 (3.3s): Aheba si Via, to Maakek Vrvrvrvrv, si Via Indmelix, arsolim, agia Ibu saidea

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 1.0000 RTF 0.99 (10.8s): ayavarsiviar stomakik veriverivriveri veri sivia in mileg al sulin aia igusaidia

Switch-point WER: 1.0000 (1 boundary)

[URGENT] WA 8.35.23 PM | Paediatrics | WhatsApp voice note (smartphone)

Reference (Sahara bootstrap): Je for a baby who is 6 months old to take water melon and eggs? Is that healthy and also soda?

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 0.1905 RTF 0.14 (1.6s): J, is it okay for a baby who is 6 months old to take water melon and eggs? Is that healthy and also soda?

Meta MMS WER 0.3333 RTF 0.93 (10.6s): j izi toke foa baby who iz sik months old to take water melon and eggs is that healthy and also sodar

[URGENT] WA 8.35.46 PM | Mental Health | WhatsApp voice note (smartphone)

Reference (Sahara bootstrap): Nasikia kichwa changu cha umma niko nimejiswa na mawazo mengi, I don't know if it's depression I don't know it's headache, I would still have some mild headache, 1-sided I don't know, we tuside unaeza.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 0.8857 RTF 0.11 (2.0s): Nesken kichachangu chawma ni kong medisana mawa zomini, I don't reach depression. I don't reach headache, Ostrilae haf sa mele headache, One

Switch-point WER: 1.0000 (1 boundary)

Meta MMS WER 0.8571 RTF 0.87 (15.9s): naskian kichwa chango cha uma nikamejiswa na mawazo mingi ivich depressionidonyvich hadikstilehave some mile hadik onesideduwe tu ni saidie

Switch-point WER: 0.8333 (1 boundary)

[URGENT] WA 8.36.04 PM | Headache & Neurology | WhatsApp voice note (smartphone)

Reference (Sahara bootstrap): No, I have a serious headache. What can I do? Please help me. Na kit.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER 1.0000 RTF 0.15 (1.5s): Kirao, I have a senior in a hindi. What can I do? Please help me. No worry, I keep trying to fight for this idea.

Switch-point WER: 0.8333 (1 boundary)

Meta MMS WER 1.0000 RTF 1.06 (10.7s): heloo aihavasiria hendek utkinaindu plees helkmi navueda kichwa sala alisaidi

Switch-point WER: 1.0000 (1 boundary)

[URGENT] WA 8.36.42 PM | Fever & Infection | WhatsApp voice note (smartphone)

Reference (Sahara bootstrap): I don't know, but I been profily sweating and yeah I was really sweating, I don't know, but I started on Saturday, like Saturday usiku nilikuwa tumelala lakini nikaanza ku sweat sijui ndio maana nimekuja niyaangalie.

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr
OpenAI Whisper WER 0.5278 RTF 0.66 (20.0s): I don't know but I've been profissively sweating. I was really sweating. I don't know but it started on Saturday. But today I'm going home w
Meta MMS WER 0.8889 RTF 0.88 (26.7s): ubtabitropetly sweting yaauredsatarde usiku mlikwa tunmelala lakini nikaanza kuswet atisijunda maana nimekuja ni aangaliliwe

[ROUTINE] WA 8.37.02 PM Chronic Disease WhatsApp voice note (smartphone)
<i>Reference (Sahara bootstrap): What is hypertension? And niliambiwa na daktari can take Panadol, paracetamol Ibuprofen. Yeah, I don't know if that is okay to take medicine when you have a headache.</i>
Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr
OpenAI Whisper WER 0.4643 RTF 0.12 (2.1s): What is hypertension? Nilemban, aecteore, kintik, panado, parasitamo, ya brufen? Ya, I don't like that is ok. To take medicine when you have
Switch-point WER: 0.6667 (1 boundary)
Meta MMS WER 0.8571 RTF 0.89 (15.3s): watiz hipe attention and niliambiwa na daktari kenteke panado parasitamo ya brufen ya edonet thatiz oke tuteke medicini wenyuhavar hedek
Switch-point WER: 0.3333 (1 boundary)

[URGENT] WA 8.38.08 PM Paediatrics WhatsApp voice note (smartphone)
<i>Reference (Sahara bootstrap): Niko na mtoto anakohoa sana usiku, like he is really coughing, but it's only at night, which is so wered. Mchana ako sawa ni makamasi kidogo na imekuwa very recurrent. Initially tulikuwa tumeambiwa we avoid far and aroun</i>
Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr
OpenAI Whisper WER 0.9273 RTF 0.66 (20.5s): Maunham'a wa' avk 1977 a runs animals, i don't know if there's anything, i niza kwa ni alaji, niza kwa bado anabir linwien on 40 zips.
Switch-point WER: 1.0000 (1 boundary)
Meta MMS WER 0.6727 RTF 0.86 (26.8s): mikona mtoto anakohoa sana usiku lekiz reli kofing bat ichonle etnit wichesouyad mchana ko sawa ni makamasi kidogo na imekuwa vere rekurrent
Switch-point WER: 0.6667 (1 boundary)

[ROUTINE] WA 8.39.00 PM Musculoskeletal WhatsApp voice note (smartphone)
<i>Reference (Sahara bootstrap): Is there anything like weak bonds ama soft bonds? Mtu akigonga mguu anasikia uchungu na ni kugongwa tu siki tunafaa kumpea uchungu alafu ukipona inakaribia inaweza kuwa nini?</i>
Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr
OpenAI Whisper WER 1.5357 RTF 0.32 (5.5s): I said anything like with bones, I'm a soft-bones. I'm talking to my husband, he's a strong dog, I'm talking to his wife, he's a strong dog.
Switch-point WER: 0.8333 (1 boundary)
Meta MMS WER 0.7143 RTF 0.85 (14.5s): isiwe initing laike wick bons ama softbons mtu wa kigungwa mguu anasikia uchungu na ni kugungwa tu hii atiki tunafaa kumpe uchungu alafu kup
Switch-point WER: 1.0000 (1 boundary)

[URGENT] WA 8.40.30 PM Respiratory WhatsApp voice note (smartphone)
<i>Reference (Sahara bootstrap): Niliamba niko na kipindupindu, alafu daktari did me some x-rays on the chest cause I was also feeling some pain like know what I can do so that I can so.</i>
Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr
OpenAI Whisper WER 0.5806 RTF 0.10 (1.7s): Nilembani kona ki pindu pindu, ala fodak teri did me some x-rays on the chest, coz I was also feeling something. Might you help me like no w
Switch-point WER: 1.0000 (1 boundary)
Meta MMS WER 0.8710 RTF 0.88 (14.6s): niliambanikona kipindupindu halafu daktary did me same ex ras on the chas kuzawazouso filing sampein amaituu helpmi lakino uwate kendu sotha
Switch-point WER: 1.0000 (1 boundary)

WA 8.41.07 PM Unknown WhatsApp voice note (smartphone)
<i>Reference (Sahara bootstrap):</i>

Intron Sahara: ERROR ? Intron API error 400: {"data":{},"message":"insufficient balance to pr

OpenAI Whisper WER ? RTF 0.82 (15.1s): Nili pata kifaba Therefore, There was difficulty being a ??? listings, eitimomou. So he regret? taoshke a' angske'n. En-takixesik It?ismane

Meta MMS WER ? RTF 0.86 (15.9s): nilipata kifafa baada ya kufanya mazoezi wchulai du izit normal ya arigressi sheking ni kaskadi kkufa io kifafa ilinsumbwa

5c. Short triage phrases (30 mp3 clips)

File	Reference (Sahara)	Sahara WER	Whisper WER	MMS WER	Best
ks_1.mp3	I need to see a doctor today.	0.0000	0.0000	0.0000	Intron
ks_2.mp3	Nahitaji kumwona daktari leo.	0.0000	0.5000	0.0000	Intron
ks_3.mp3	Nahitaji kuona doc leo.	API err	0.7500	0.2500	Meta
ks_4.mp3	My head hurts badly.	API err	0.0000	0.5000	OpenAI
ks_5.mp3	Kichwa kinaniuma sana.	API err	1.3333	0.6667	Meta
ks_6.mp3	Kichwa inauma mbaya.	API err	1.0000	0.0000	Meta
ks_7.mp3	Remember to take your medicine.	API err	1.0000	0.0000	Meta
ks_8.mp3	Kumbuka kumeza dawa zako.	API err	0.7500	0.0000	Meta
ks_9.mp3	Kumbuka kumeza dawa zako za pr..	API err	0.8333	0.3333	Meta
ks_10.mp3	I am feeling much better now.	API err	0.0000	0.0000	OpenAI
ks_11.mp3	Najisikia vizuri zaidi sasa.	API err	1.0000	0.0000	Meta
ks_12.mp3	Najisikia fit sana saa hii.	API err	1.0000	0.2000	Meta
ks_13.mp3	Washi your hands with soap.	API err	1.0000	0.4000	Meta
ks_14.mp3	Osha mikono yako kwa sabuni.	API err	1.0000	0.0000	Meta
ks_15.mp3	Osha mikono yako na soap.	API err	0.8000	0.2000	Meta
ks_16.mp3	He has a severe cough.	API err	1.0000	0.4000	Meta
ks_17.mp3	Anakohoa sana.	API err	1.0000	0.0000	Meta
ks_18.mp3	Ana cough mbaya sana.	API err	1.0000	0.2500	Meta
ks_19.mp3	Drink plenti of water everyday.	API err	1.2000	0.2000	Meta
ks_20.mp3	Kunywa maji mengi kila siku.	API err	1.0000	0.0000	Meta
ks_21.mp3	Kunywa maji mengi kila day.	API err	1.0000	0.2000	Meta
ks_22.mp3	The clinic is closed on sundays.	API err	0.6667	0.0000	Meta
ks_23.mp3	Sahanati imefungwa siku ya Jum..	API err	1.4000	0.2000	Meta
ks_24.mp3	Hiyo clinic imefungwa Sunday.	API err	1.2500	0.2500	Meta
ks_25.mp3	Where is the nearest hospital?	API err	1.0000	0.0000	Meta
ks_26.mp3	Hospitali ya karibu iko wapi?	API err	0.8000	0.0000	Meta
ks_27.mp3	Hospital nearest iko wapi?	API err	0.7500	0.7500	OpenAI
ks_28.mp3	Can I book an appointment?	API err	1.0000	0.2000	Meta
ks_29.mp3	Naweza kuweka miadi.	API err	1.0000	0.0000	Meta
ks_30.mp3	Naweza kubuka appendicid.	API err	0.3333	0.3333	OpenAI

SECTION 6 ? Known Limitations

1. NO HAND-WRITTEN GROUND TRUTH

Sahara bootstrap transcripts are used as the reference. This makes Sahara WER artificially low (near 0) and means Whisper and MMS WER reflects disagreement with Sahara, not actual transcription errors. Manual transcription of at least the m4a and ogg files is needed for a true accuracy benchmark.

2. SAHARA API BALANCE RAN OUT

The Intron Sahara API returned an 'insufficient balance' error part-way through the benchmark run. Sahara results are marked 'API err' for affected clips. Replenish the API balance and re-run to get Sahara results for all 65 clips.

3. SINGLE SPEAKER IN m4a SET

All 20 m4a clips are recordings by one person. Results on this subset do not reflect how well the system works with different voices, accents or ages.

4. TRIAGE LABEL MATCHING

Intent accuracy uses exact string matching. The triage engine and the annotation often describe the same condition in different words. This inflates the apparent error rate. A fuzzy or category-based match would be fairer.

5. LOCAL MODEL SPEED

Whisper and MMS timing is CPU-only. RTF would drop significantly on a machine with a GPU. Numbers here are for reference only and depend on the test machine.

6. QUIET RECORDING CONDITIONS

Most clips were recorded in quiet rooms. Real clinics have background noise, multiple speakers, and variable mic quality. The ogg WhatsApp clips (15 clips) have some ambient noise and are more representative of real conditions.